



A lognormal-normal mixture model for unsupervised health indicator construction and its application into gear remaining useful life prediction

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ABSTRACT

Accurately predicting the remaining useful life (RUL) of a key component, such as gear, is significant for guaranteeing the safe operation of mechanical equipment and making a proper maintenance plan. The health indicator (HI) plays an essential role in the data-driven RUL prediction technique. HI can be constructed from the perspective of the data distribution discrepancy. However, some existing methods cannot utilize different types of distributions to estimate the distribution discrepancy in various domains. In addition, the constructed HI may not comprehensively describe the tendency of performance degradation by using a type of distribution to obtain the distribution discrepancy in a domain. To overcome these challenging problems, a novel lognormal-normal mixture model (LNMM) that utilizes lognormal and normal distributions is constructed to estimate data distributions from two data domains, including the raw data domain and exponentially transformed data domain. Then, the distribution contact ratio metric (DCRM) is applied to calculate the discrepancies between benchmark distribution of healthy data and distributions of whole life-cycle data in two domains. The gear HI is generated without supervision by combining the DCRMs of two domains. The developed unsupervised HI is employed to estimate gear's RUL via an improved multi-hierarchical long-term memory augmented network. Finally, the experimental results indicate the feasibility and merit of the developed LNMM in gear HI construction. The LNMM-based HI has a better predictive efficacy than the conventional and state-of-the-art unsupervised HIs.

1. Introduction

As the essential transmission component in rotating machinery, gears have been extensively employed in transmission systems of multiple equipment, including aero-engines, mining machineries, sophisticated machine tools, and wind turbines [1,2]. Gears are prone to failure in practical application due to the adverse working environment and prolonged high-load operation. As operating time increases, different faults may emerge on the gear tooth surface, such as wear, fracture, and broken teeth [3,4]. To guarantee the risk-free operation of machinery and avoid the aforementioned gear faults, it is imperative to implement precise and dependable remaining useful life (RUL) prediction. Some preventive maintenance plans can be made based on the predicted RUL to reduce downtime and

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maintenance costs or even prevent some serious catastrophic accidents [5–7]. Thus, it is critically indispensable to perform RUL prediction for machine.

As the pivotal technology in the prognostics and health management, the RUL prediction methods can be roughly classified into physics-based and data-driven methods [8,9]. The physics-based approach is developed to identify the degeneration process of machine by constructing a complicated physical model. Accurate RUL can be calculated using this physical model. However, it is extremely hard to design an appropriate physical model for complicated mechanical equipment in real industry. The data-driven approach is explored for achieving life prediction of complex machinery, extracting degradation information from monitoring data through machine learning or deep learning models. The data-driven approach can automatically extract abstract features from complex monitoring data with minimal reliance on prior knowledge and human intervention. In recent years, some advanced data-driven methods have been developed for the RUL prediction of mechanical equipment [10,11]. Cao et al. [12] presented a probabilistic life prediction approach employing recurrent neural network (RNN) and attention mechanism to achieve precise life estimation for wind turbine bearings. Chen et al. [13] introduced a data-driven methodology aimed at predicting the RUL of a machine. The significance of features and time steps can be effectively learned via an attention mechanism. Xiang et al. [14] put forward a multi-cellular long short-term memory (LSTM) for RUL prediction of aero-engines. Chen et al. [15] built a new variant of RNN model to estimate the RULs of servo motors. This model can learn the importance of the outputs of two gates and fully make use of the hierarchical information of network. Ma and Mao [16] constructed a convolution-based LSTM model to estimate machine's RUL by leveraging the mined information from the monitoring data. These existing data-driven methods can obtain accurate RUL prediction results based on RNN model or attention mechanism. However, the long-term memory capacity based on gating mechanism of RNN is ignored in these methods. This will result in the loss of some crucial degradation information during the process of RUL prediction.

The existing data-driven RUL prediction framework can be segmented into several stages: monitoring data acquisition, health indicator (HI) construction (feature extraction), and RUL prediction (regression method or time-series method). In the initial stage, different types of monitoring data are obtained by various sensors. During the second stage, the acquired monitoring data is processed to generate a favorable HI that effectively reflects the health condition of the mechanical equipment. With the constructed HI, the machine's RUL can be precisely estimated via the life prediction methods in the last stage. Within the outlined data-driven RUL prediction framework, the HI construction stage is significant for the data-driven RUL estimation method. An appropriate HI can evaluate the entire degradation process of machinery and guarantee the accurate prediction results of prediction approaches. Therefore, it is essential to explore a novel type of HI construction method to build HI in quantitatively evaluating the deterioration trajectory of machinery and obtaining accurate and reliable estimation results.

In recent years, extensive research efforts have been dedicated to exploring various approaches for constructing HIs. These constructed HIs contain physical HIs and virtual HIs. For the physical HIs, the statistic or signal processing methods are applied to construct HIs from the acquired run-to-failure signals. Huang et al. [17] utilized root mean square (RMS) to predict the bearing's RUL via an improved stochastic model. Li et al. [18] introduced a novel HI to evaluate the degradation process of bearing using the mathematical morphology theory. Soualhi et al. [19] constructed a new physical HI from raw bearing's vibration signals via the Hilbert-Huang transform. Javed et al. [20] built two physical HIs using trigonometric functions, and their monotonicity was improved via an accumulation strategy. Wang et al. [21] designed a novel HI based on energy evolution to evaluate the degradation of a pump impeller. According to the above works about physical HIs, they can assess the degradation process of mechanical component to some extent. However, the physical HIs based on manually extracted features cannot provide a comprehensive depiction of the deterioration process of a mechanical equipment.

Hence, the virtual HI construction approach has been proposed to address this issue and constructs abstract HIs from various physical features or raw vibration signals using dimension reduction approaches or neural network methods. These methods can automatically extract HI without any expert knowledge and human labor. Wu et al. [22] explored a principal component Mahalanobis distance HI by fusing multiple degradation features with dynamic principal component analysis and Mahalanobis distance. Li et al. [23] introduced a novel HI construction model for improving the monotonicity of the constructed HI. He and Guo [24] presented an unsupervised HI construction method based on Gaussian mixture mode (GMM) to generate HIs of bearings. Ni et al. [25] proposed a new virtual HI to capture the degradation progression of bearing over time. In addition, this HI is capable of eliminating random fluctuation. Chen et al. [26] developed a performance degradation assessment framework to generate bearing HI based on an auto-encoder network and growing self-organizing map. Li et al. [27] used an improved adversarial network to construct bearing HI with only normal data. The constructed HI could evaluate the bearing's degradation trajectory accurately and monotonously. Zhou et al. [28] introduced a novel unsupervised HI construction approach to construct a bearing's HI from vibration signals with GMM and the proposed distribution contact ratio metric (DCRM). The existing literatures about HI construction imply that the unsupervised clustering method is effective for constructing HI from the monitoring data, especially for GMM. The GMM method can estimate the unknown distributions of life-cycle data by combining several normal distributions. Based on the estimated distributions, HI is constructed via some distance metrics, which can measure the distribution discrepancies between distribution of health sample and distributions of the run-to-failure samples.

However, the GMM estimates distribution from the perspective of normal distribution in a data domain. The distribution characteristics in a single domain cannot provide a complete representation of machine's health condition and degradation trend. Thus, it is imperative to capture the additional degradation characteristics from another data domain. The new degradation characteristics can be generated by transforming the raw data into those of another domain via nonlinear transformation methods. As a common data nonlinear transformation method, exponential transformation is often utilized to deal with the nonlinear relationship of data by transforming it into exponential data domain. After exponential transformation, the obtained data are not subjected to the normal distribution. Their distribution characteristics cannot be estimated using GMM. Therefore, a new mixture model that considers

different types of distributions should be explored to address this issue by simultaneously estimating the data distributions from two data domains, including the raw data domain and the exponentially transformed data domain.

According to a series of tests, it is found that the mechanical vibration signals are generally subject to normal distribution [29,30], and the corresponding exponentially transformed signals mainly contain lognormal distribution features. Therefore, a novel lognormal-normal mixture model (LNMM) is designed to synchronously extract the distribution characteristics from the raw data domain and the exponentially transformed data domain. Then, the DCRM is employed to generate the gear's HI by measuring the distribution discrepancies between the benchmark distribution of health data and the distributions of the full life-cycle data. Since the lognormal and normal distribution characteristics are simultaneously extracted by the proposed LNMM from two data domains, the proposed HI based on LNMM and DCRM can comprehensively represent the gear's degradation trend from various aspects. The RULs of gears can be predicted accurately and robustly using the LNMM-based HIs and an improved multi-hierarchical long-term memory augmented network (IMLMA-Net). The advantage of the developed HI construction approach based on LNMM is validated by two gear life-cycle datasets. The following are the main contributions of this work:

- (1) The lognormal mixture model (LMM) is first developed to estimate the data distribution of the vibration signals after exponential transformation. Based on GMM and LMM, the LNMM is constructed to simultaneously estimate data distributions from two various data domains, including raw vibration signals and data after exponential transformation. The corresponding parameter updating formulas of LMM and LNMM are also derived.
- (2) A novel unsupervised HI construction approach is put forward for describing the tendency of performance degradation via the developed LNMM. With the estimated distribution parameters by LNMM, the gear's HI is constructed by using DCRM to measure the distribution discrepancies between the benchmark distribution of health signal and distributions of the entire life-cycle signals.
- (3) The IMLMA-Net is developed to enhance the memory capacity of MLMA-Net, and it is used for RUL prediction with the proposed HI. The experimental results on several gear life-cycle datasets verify the feasibility and merit of the presented LNMM in gear HI construction. Lastly, the LNMM-based HI is more appreciate for RUL prediction than the classical and advanced HIs.

The remainder of this article is structured as follows. Section 2 details the methodology of the LNMM in constructing HI from the perspective of data distribution discrepancy. Section 3 demonstrates the feasibility and merit of the proposed LNMM using several gear life-cycle datasets, and illustrates the application of LNMM-based HI in life prediction. Finally, Section 4 concludes this article.

2. The proposed HI construction methodology

In this section, the LMM is first developed to estimate data distribution from the perspective of lognormal distribution in the exponential data domain. Then, the LNMM is proposed based on GMM and LMM to extract the distribution characteristics from two domains. Next, the proposed HI construction method based on LNMM is developed for generating gear HI from the perspective of distribution discrepancies. Finally, the gear RUL prediction method based on IMLMA-Net is introduced for RUL prediction.

2.1. Lognormal mixture model

The GMM has been used to estimate many unknown data distributions. However, not all data distributions can be estimated via GMM. It is mainly applicable to continuous random variables by assuming that the data distribution is a linear combination of a series of normal distributions. However, if the distribution characteristics in a single data domain are quite different from normal distribution characteristics, GMM may not provide favorable estimated results. For example, when the data is transformed by a nonlinear transformation method (e.g., exponential transform), the distribution characteristics of the transformed data cannot be properly extracted via GMM. This is because that the distribution characteristics of the transformed data change and have a major discrepancy from the normal distribution characteristics. Thus, the LMM is introduced in this work for estimating the data distribution from the exponential data domain. As a variant of GMM, LMM is able to estimate the data distribution from the perspective of lognormal distribution in a data domain. The overall density function of LMM is formulated as

$$P(x|\theta) = \sum_{l=1}^L \alpha_l \phi(x|\theta_l), \quad (1)$$

where $\phi(x|\theta_l) = \frac{1}{\sqrt{2\pi}\sigma_l x} \exp\left(-\frac{(\log x - \mu_l)^2}{2\sigma_l^2}\right)$, $\sum_{l=1}^L \alpha_l = 1, \alpha_l > 0$. According to Eq. (1), it can be found that the normal density function in GMM is replaced by the lognormal density function in LMM.

Subsequently, the estimation of LMM parameters is conducted employing the EM method. The observed data is $x = (x_1, x_2, \dots, x_M)$. A latent variable γ_{jl} is introduced to denote that the observed data x_j comes from the l th component in LMM, which is defined as follows:

$$\gamma_{jl} = \begin{cases} 1, & x_j \text{ comes from the } l\text{th component} \\ 0, & \text{Otherwise} \end{cases}, \quad j = 1, 2, \dots, M; \quad l = 1, 2, \dots, L, \quad (2)$$

where γ_{jl} is 0–1 random variable. With the observed data x_j and unobserved data γ_{jl} , the complete data is obtained as $(x_j, \gamma_{j1}, \gamma_{j2}, \gamma_{j3}, \dots, \gamma_{jL}), j = 1, 2, \dots, M$. Thereupon, the likelihood function of the complete data is written as follows:

$$\begin{aligned} P(\mathbf{x}, \gamma | \theta) &= \prod_{j=1}^M P(x_j, \gamma_{j1}, \gamma_{j2}, \dots, \gamma_{jL} | \theta), \\ &= \prod_{l=1}^L \prod_{j=1}^M [\alpha_l \phi(x_j | \theta)]^{\gamma_{jl}} \\ &= \prod_{l=1}^L \alpha_l^{n_l} \prod_{j=1}^M [\phi(x_j | \theta)]^{\gamma_{jl}} \\ &= \prod_{l=1}^L \alpha_l^{n_l} \prod_{j=1}^M \left[\frac{1}{\sqrt{2\pi} \sigma_l x_j} \exp \left(-\frac{(\log x_j - \mu_l)^2}{2\sigma_l^2} \right) \right]^{\gamma_{jl}} \end{aligned} \quad (3)$$

where $n_l = \sum_{j=1}^M \gamma_{jl}$, $\sum_{l=1}^L n_l = M$. Then, the log-likelihood function of the complete data is expressed as follows:

$$\log P(\mathbf{x}, \gamma | \theta) = \sum_{l=1}^L \left\{ n_l \log \alpha_l + \sum_{j=1}^M \gamma_{jl} \left[\log \left(\frac{1}{\sqrt{2\pi}} \right) - \log \sigma_l - \log x_j - \frac{1}{2\sigma_l^2} (\log x_j - \mu_l)^2 \right] \right\}. \quad (4)$$

In the subsequent step of the EM algorithm, referred to as the E-step, the Q-function is calculated, defined as follows:

$$\begin{aligned} Q(\theta, \theta^{(i)}) &= E[\log P(\mathbf{x}, \gamma | \theta) | \mathbf{x}, \theta^{(i)}] \\ &= E \left\{ \sum_{l=1}^L \left\{ n_l \log \alpha_l + \sum_{j=1}^M \gamma_{jl} \left[\log \left(\frac{1}{\sqrt{2\pi}} \right) - \log \sigma_l - \log x_j - \frac{1}{2\sigma_l^2} (\log x_j - \mu_l)^2 \right] \right\} \right\}, \\ &= \sum_{l=1}^L \left\{ \sum_{j=1}^M E(\gamma_{jl}) \log \alpha_l + \sum_{j=1}^M E(\gamma_{jl}) \left[\log \left(\frac{1}{\sqrt{2\pi}} \right) - \log \sigma_l - \log x_j - \frac{1}{2\sigma_l^2} (\log x_j - \mu_l)^2 \right] \right\} \end{aligned} \quad (5)$$

where $E(\gamma_{jl} | \mathbf{x}, \theta)$ needs to be calculated based on current model parameters denoted as $\hat{\gamma}_{jl}$:

$$\begin{aligned} \hat{\gamma}_{jl} &= E(\gamma_{jl} | \mathbf{x}, \theta) = P(\gamma_{jl} = 1 | \mathbf{x}, \theta) \\ &= \frac{P(\gamma_{jl} = 1, x_j | \theta)}{P(x_j | \theta)} = \frac{P(\gamma_{jl} = 1, x_j | \theta)}{\sum_{l=1}^L P(\gamma_{jl} = 1, x_j | \theta)} \\ &= \frac{P(x_j | \gamma_{jl} = 1, \theta) P(\gamma_{jl} = 1 | \theta)}{\sum_{l=1}^L P(x_j | \gamma_{jl} = 1, \theta) P(\gamma_{jl} = 1 | \theta)} = \frac{\alpha_l \phi(x_j | \theta_l)}{\sum_{l=1}^L \alpha_l \phi(x_j | \theta_l)} \end{aligned} \quad (6)$$

After that, the Q-function can be expressed as follows:

$$\begin{aligned} Q(\theta, \theta^{(i)}) &= \sum_{l=1}^L \left\{ \sum_{j=1}^M \hat{\gamma}_{jl} \log \alpha_l + \sum_{j=1}^M \hat{\gamma}_{jl} \left[\log \left(\frac{1}{\sqrt{2\pi}} \right) - \log \sigma_l - \log x_j - \frac{1}{2\sigma_l^2} (\log x_j - \mu_l)^2 \right] \right\}, \\ &= \sum_{l=1}^L \sum_{j=1}^M \left\{ \hat{\gamma}_{jl} \log \alpha_l + \hat{\gamma}_{jl} \left[\log \left(\frac{1}{\sqrt{2\pi}} \right) - \log \sigma_l - \log x_j - \frac{1}{2\sigma_l^2} (\log x_j - \mu_l)^2 \right] \right\} \end{aligned} \quad (7)$$

The M-step of EM is used to compute the model parameters for the next iteration by maximizing the Q-function:

$$\theta^{(i+1)} = \underset{\theta}{\operatorname{argmax}} Q(\theta, \theta^{(i)}). \quad (8)$$

The optimal $\hat{\mu}_l$, $\hat{\sigma}_l^2$, and $\hat{\alpha}_l$ are found by equating the perspective partial derivatives with zero and solving for $\hat{\mu}_l$ to obtain:

$$\hat{\mu}_l = \sum_{j=1}^M \hat{\gamma}_{jl} \log x_j / \sum_{j=1}^M \hat{\gamma}_{jl}. \quad (9)$$

For $\hat{\sigma}_l^2$, the following expression is obtained:

$$\hat{\sigma}_l^2 = \sum_{j=1}^M \hat{\gamma}_{jl} (\log x_j - \hat{\mu}_l)^2 / \sum_{j=1}^M \hat{\gamma}_{jl}. \quad (10)$$

For $\hat{\alpha}_l$, the following expression is obtained:

$$\hat{\alpha}_l = \sum_{j=1}^M \hat{\gamma}_{jl} / M. \quad (11)$$

The parameter estimation procedure of LMM is shown in Table 1.

2.2. Lognormal-normal mixture model

The GMM or LMM can only estimate the data distribution in a data domain from the perspective of normal or lognormal distributions. However, the distribution characteristics in a single domain cannot completely represent the health status and degradation process of machinery. Thus, it is essential to explore a novel mixture model to mine additional degradation characteristics from two data domains. In this work, the LNMM is cautiously designed to estimate data distributions from two data domains, including raw and exponentially transformed data. Normal and lognormal distributions are both considered in LNMM. The overall density function of LNMM is defined as follows:

$$P(x, x' | \theta) = \sum_{k=1}^K \alpha_k \varphi(x | \theta_k) + \sum_{l=1}^L \alpha_l \phi(x' | \theta_l), \quad (12)$$

where x is the raw data, x' is the exponentially transformed data; K and L represent the number of normal and lognormal distributions, respectively, α is the positive weight coefficient that satisfies $\sum_{k=1}^K \alpha_k + \sum_{l=1}^L \alpha_l = 1$, and $\theta_k = (\mu_k, \sigma_k^2)$, $\theta_l = (\mu_l, \sigma_l^2)$.

The EM algorithm is applied for parameter estimation of LNMM. In the E step, the responsiveness of the component k to the raw data x_j is calculated as follows:

$$\hat{\gamma}_{jk} = \alpha_k \varphi(x_j | \theta_k) / \left(\sum_{k=1}^K \alpha_k \varphi(x_j | \theta_k) + \sum_{l=1}^L \alpha_l \phi(x'_j | \theta_l) \right), j = 1, 2, \dots, M; k = 1, 2, \dots, K; l = 1, 2, \dots, L. \quad (13)$$

Furthermore, the responsiveness of the component l to the exponentially transformed data x'_j is obtained via the following formula:

$$\hat{\gamma}_{jl} = \alpha_l \phi(x'_j | \theta_l) / \left(\sum_{k=1}^K \alpha_k \varphi(x_j | \theta_k) + \sum_{l=1}^L \alpha_l \phi(x'_j | \theta_l) \right), j = 1, 2, \dots, M; k = 1, 2, \dots, K; l = 1, 2, \dots, L. \quad (14)$$

Based on the calculated responsiveness, the Q-function can be expressed in the following formula:

$$Q(\theta, \theta^{(i)}) = \sum_{k=1}^K \sum_{j=1}^M \left\{ \hat{\gamma}_{jk} \log \alpha_k + \hat{\gamma}_{jk} \left[\log \left(\frac{1}{\sqrt{2\pi}} \right) - \log \sigma_k - \frac{1}{2\sigma_k^2} (x_j - \mu_k)^2 \right] \right\} \\ + \sum_{l=1}^L \sum_{j=1}^M \left\{ \hat{\gamma}_{jl} \log \alpha_l + \hat{\gamma}_{jl} \left[\log \left(\frac{1}{\sqrt{2\pi}} \right) - \log \sigma_l - \log x'_j - \frac{1}{2\sigma_l^2} (\log x'_j - \mu_l)^2 \right] \right\}. \quad (15)$$

During the M-step, iterative update of the model parameters is achieved by maximizing the Q-function through the application of the following formulas:

$$\hat{\mu}_k = \sum_{j=1}^M \hat{\gamma}_{jk} x_j / \sum_{j=1}^M \hat{\gamma}_{jk}, \quad (16)$$

$$\hat{\sigma}_k^2 = \sum_{j=1}^M \hat{\gamma}_{jk} (x_j - \hat{\mu}_k)^2 / \sum_{j=1}^M \hat{\gamma}_{jk}, \quad (17)$$

Table 1

The parameter estimation procedure of LMM.

Algorithm 1: Parameter estimation process of LMM	
Input:	The observed data $x = (x_1, x_2, \dots, x_M)$, the constructed LMM.
Output:	The parameters of LMM.
1.	The parameters of LMM are first initialized randomly.
2. E step:	According to the current model parameters, the responsiveness of the component l to the observed data x_j is calculated according to Eq. (6).
3. M step:	Update the model parameters for the next iteration according to Eqs. (9), (10), and (11).
4.	Continue with steps 2 and 3 iteratively until reaching convergence.

$$\hat{\alpha}_k = \sum_{j=1}^M \hat{\gamma}_{jk} / M, \quad (18)$$

$$\hat{\mu}_l = \sum_{j=1}^M \hat{\gamma}_{jl} \log x'_j / \sum_{j=1}^M \hat{\gamma}_{jl}, \quad (19)$$

$$\hat{\sigma}_l^2 = \sum_{j=1}^M \hat{\gamma}_{jl} (\log x'_j - \hat{\mu}_l)^2 / \sum_{j=1}^M \hat{\gamma}_{jl}, \quad (20)$$

$$\hat{\alpha}_l = \sum_{j=1}^M \hat{\gamma}_{jl} / M. \quad (21)$$

The parameter estimation procedure of LNMM is shown in Table 2.

2.3. Proposed HI construction method

There are three primary stages in the developed HI construction approach. In the first stage, the monitoring data of gear are acquired via several accelerometers. Noises in the original life-cycle data are preprocessed and reduced. The LNMM is carefully designed in the second stage, and data distributions are estimated by the constructed LNMM from two data domains. In the last stage, the DCRM is used to generate the gear's HI based on the estimated benchmark distribution of health data and distributions of the entire life-cycle data. The flowchart illustrating the presented HI construction methodology is depicted in Fig. 1.

The specific procedures of the developed HI construction approach are summarized as follows:

(1) Monitoring Data Acquisition: The gear's vibration signals are collected via the installed accelerometers under specific working conditions. Then, raw vibration signals are processed by the noise reduction method. The processed monitoring data is $\mathbf{X} = (\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_M)^T$, $\mathbf{x}_i = (x_{i1}, x_{i2}, \dots, x_{im})$, where M is the total number of the sampled signals and m is the data length of a single sampled signals.

(2) Data Distribution Estimation: The number of normal components and lognormal components are respectively set as K and L , and then the LNMM is constructed. In the real application, the number of normal and lognormal component can be set by Bayesian information criterion. Next, the first N healthy samples are utilized to estimate the benchmark distribution P_0 , while the distributions $\mathbf{P} = (P_1, P_2, \dots, P_M)$ of the life-cycle samples $\mathbf{X}$ are separately estimated via the presented LNMM.

(3) Gear HI Construction: The DCRM value H_i [28] is calculated based on the benchmark distribution P_0 and the estimated data distributions $\mathbf{P} = (p_1, p_2, \dots, p_M)$:

$$DCRM(P_0 \| P_i) = H_i = 1 - S_i, \quad i = 1, 2, \dots, M, \quad (22)$$

where S_i is the area of the overlapping region between P_0 and P_i . It can be obtained by the following formula:

$$S_i = \int \min[P_0(x), P_i(x)] dx. \quad (23)$$

Finally, the gear's HI vector $\mathbf{H} = (H_1, H_2, \dots, H_M)$ is generated.

2.4. Gear RUL prediction method based on IMLMA-Net

Once the HI vector is constructed using the monitoring dataset of gear, it is essential to chose a suitable RUL prediction method to estimate the gear's RUL. Essentially, the time-series-based life prediction method is able to estimate unknown HI points by learning degradation information from known HI points. In this method, the LSTM network [31] has been broadly utilized for machine's life prediction. Moreover, some variants of LSTM have been explored to enhance the memory capacity of network. In particular, the multi-hierarchical long-term memory augmented network (MLMA-Net) is capable of realizing long-term RUL prediction using the multi-hierarchical mechanism and memory augmentation function (MAF). Furthermore, the MLMA-Net has been successfully employed

Table 2

The parameter estimation procedure of LNMM.

Algorithm 2: Parameter estimation process of LNMM
Input: The raw data and exponentially transformed data, the constructed LNMM. Output: The parameters of LNMM. 1. The parameters of LNMM are first initialized randomly. 2. E step: According to the current model parameters, the responsiveness of components is calculated according to Eqs. (13) and (14). 3. M step: Update the model parameters for the next iteration according to Eqs. (16) to (21). 4. Continue with steps 2 and 3 iteratively until reaching convergence.

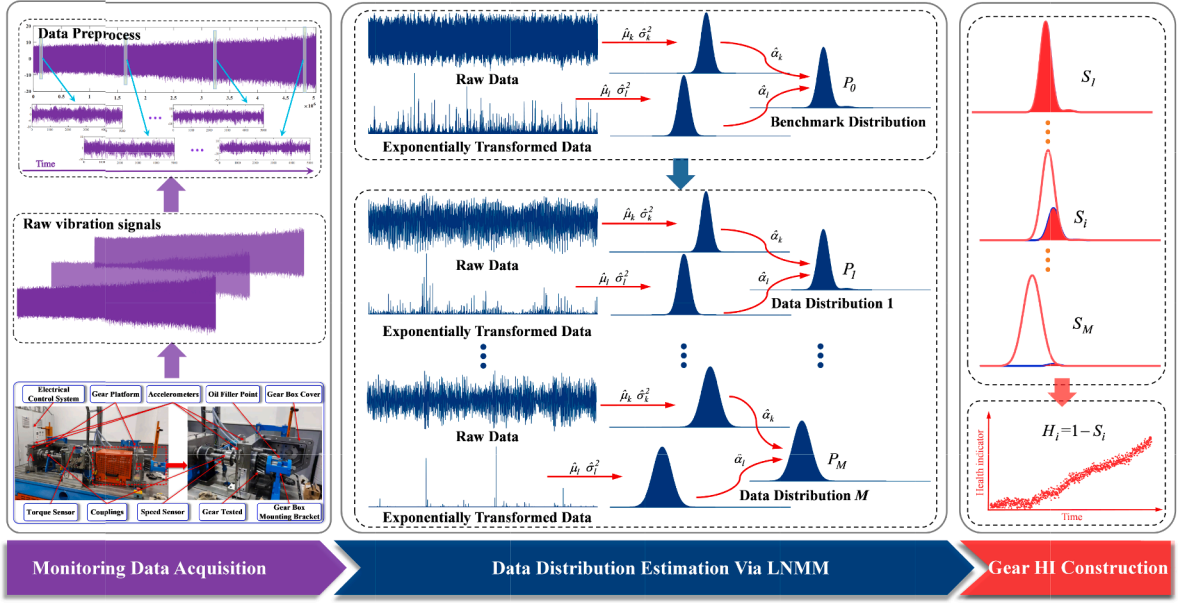


Fig. 1. The whole flowchart of the presented HI construction method.

to estimate the gear's RUL. Therefore, the MLMA-Net is chosen as the prediction network to demonstrate that the gear's HI generated by the LNMM is more appropriate for estimating gear's RUL. Detailed information about MLMA-Net can be found in [32]. However, the MAF employed in MLMA-Net might not effectively generate more gated output values in boundary regions. To address this limitation, the Gumbel function is introduced as a substitute for the MAF in MLMA-Net, which is defined as follows:

$$R(s, \eta) = \sigma\left(\frac{s + \log K - \log(1 - K)}{\eta}\right), \quad (24)$$

where s denotes the input of this function, η denotes a predetermined coefficient greater than 0, in vector K , the values are independently sampled from a uniform distribution between 0 and 1, σ denotes the sigmoid function. The Gumbel function can be used to approximate some sampled values of discrete distribution (e.g., Bernoulli distribution). An improved MLMA-Net (IMLMA-Net) is designed for RUL prediction using the Gumbel function. Fig. 2 visually represents the cell structure of IMLMA-Net, and the Gumbel function is marked by red dashed box.

Based on the cell state h_t of the IMLMA-Net, the output of the IMLMA-Net can be calculated as follows [32]:

$$\begin{aligned} y_t^1 &= \text{Dropout}(h_t) \\ y_t^2 &= \text{Dropout}(\text{Relu}(\mathbf{W}_1 y_t^1 + \mathbf{b}_1)), \\ y_t^3 &= \mathbf{W}_2 y_t^2 + \mathbf{b}_2 \\ y_t &= \mathbf{W}_3 (\text{Dropout}(y_t^3 + h_t)) + \mathbf{b}_3 \end{aligned} \quad (25)$$

where y_t is the output value of the IMLMA-Net.

The constructed HIs of gears are utilized as the input data of the IMLMA-Net for predicting gear RUL. The learnable parameters of IMLMA-Net are initialized using a uniform initialization method and subsequently updated via the stochastic gradient descent (SGD) algorithm. The prediction flowchart is depicted in Fig. 3.

3. Experimental results and comparison

The monitoring data of gears from two test rigs are acquired for HI construction. The gear's HIs are generated via the proposed LNMM to quantitatively describe the degradation processes of gears. Several traditional and advanced HI construction methods are utilized to validate the effectiveness and superiority of LNMM by comparing them with the proposed LNMM using the evaluation metrics. The gear's RUL is predicted via the extracted LNMM-based HIs and the constructed IMLMA-Net to prove that the LNMM-based HI is appropriate for RUL prediction. The advantage of the LNMM-based HI for life prediction is validated through a comparative analysis with other HIs.

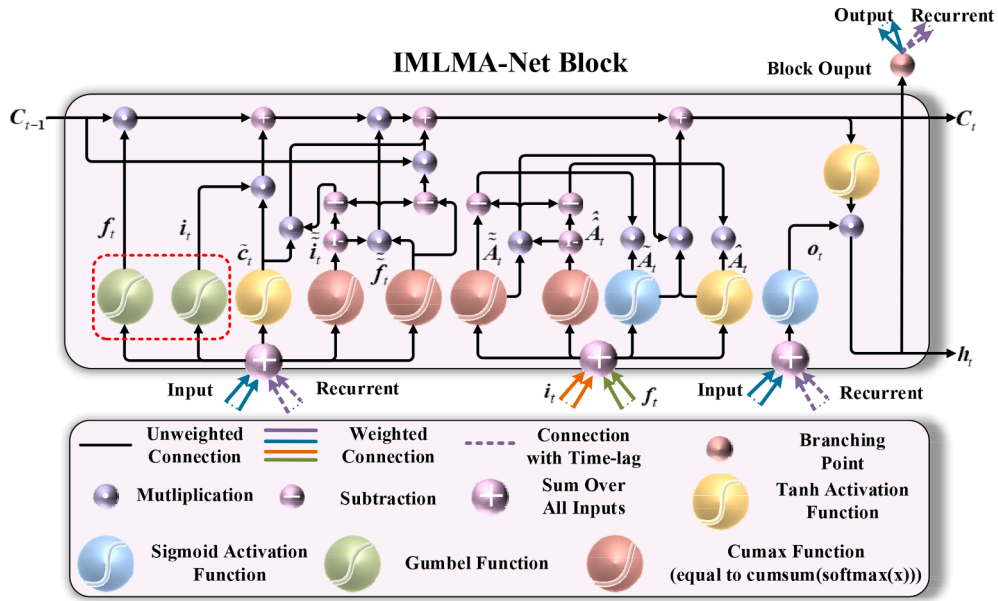


Fig. 2. The cell structure of IMLMA-Net.

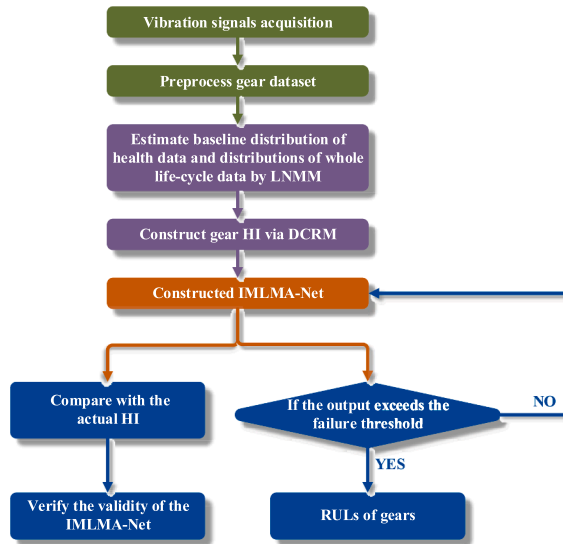


Fig. 3. The prediction flowchart for RUL prediction.

3.1. Experiment description

The vibration signals of gear dataset ¹ are acquired by a gear contact fatigue testing platform at Newcastle University, UK. The testing platform, illustrated in Fig. 4, includes an electrical control system, a gear platform, four accelerometers, four couplings, a torque sensor, and a speed sensor. The gear platform has two gearboxes (e.g., gearbox A and gearbox B), enabling simultaneous testing of gears in both gearboxes under identical operating conditions.

During the gear life test, the vibration signals from gearboxes A and B were acquired using installed accelerometers. The sampling frequency and sampling duration were configured at 25 kHz and 0.2 s, respectively, with a sampling interval of 1 min. The terminating condition for the test was when the amplitude of the recorded data exceeded the preset threshold. The life-cycle datasets of gears under two distinct operating conditions were obtained, as shown in Table 3.

¹ <https://qinyi-team.github.io/2024/01/Gear-life-cycle-data-download/>.

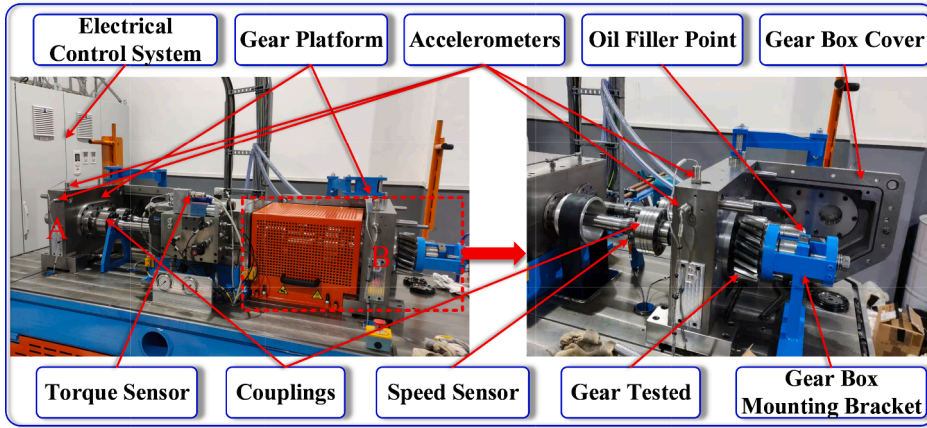


Fig. 4. Gear contact fatigue testing platform for gear dataset 1.

Table 3

Data description of gear dataset 1.

Operating condition	Condition1	Condition2
Rotating speed (rpm)	1100	800
Torque (Nm)	4400	4070
Gearbox A	GearA1-1	GearA2-1
	GearA1-2	GearA2-2
Gearbox B	GearB1-1	GearB2-1
	GearB1-2	GearB2-2

Three run-to-failure datasets of gear dataset 2² were acquired via another gear contact fatigue testing platform manufactured by Strama-MPS Inc [33], as shown in Fig. 5. This platform comprises an operating platform, a gear test platform, a torque control system, and a hydraulic cooling system. The vibration signals were recorded during the life test using embedded accelerometers, with a sampling frequency of 50 kHz, a sampling duration of 10 s, and a sampling interval of 1 min. The test was finished when the amplitude of the vibration signal surpassed the predetermined threshold. The collected gear datasets are described in Table 4. The vibration signals collected during the steady stage exhibited a high degree of consistency. As such, they were not particularly useful for life prediction. Thus, only the final 600 samples were utilized in this work.

3.2. Evaluation metrics

Three common metrics, monotonicity (*Mon*), correlation (*Cor*), and robustness (*Rob*), are applied to quantitatively evaluate the performance of the constructed gear HI. The obtained HI vector is first decomposed into an average trend and a random term:

$$H(t_m) = H_T(t_m) + H_R(t_m), \quad (26)$$

where $H(t_m)$ is the HI value at time t_m , $H_T(t_m)$ is the average trend, and $H_R(t_m)$ is the random term.

Based on the acquired average trend H_T and random term H_R , *Mon*, *Cor*, and *Rob* are calculated by the following formulas [34]:

$$Mon(H) = \left| \frac{No.ofdH_T > 0}{M-1} - \frac{No.ofdH_T < 0}{M-1} \right|, \quad (27)$$

$$Cor(H, T) = \frac{|\sum_{m=1}^M (H(t_m) - \bar{H}) \times (t_m - \bar{T})|}{\sqrt{\sum_{m=1}^M (H(t_m) - \bar{H})^2 \times \sum_{m=1}^M (t_m - \bar{T})^2}}, \quad (28)$$

$$Rob(H) = \exp\left(-\frac{\lambda}{M} \sum_{m=1}^M \left| \frac{H_R(t_m)}{\max(H) - \min(H)} \right| \right), \quad (29)$$

² <https://qinyi-team.github.io/2019/12/Gear-life-cycle-data-download/>.

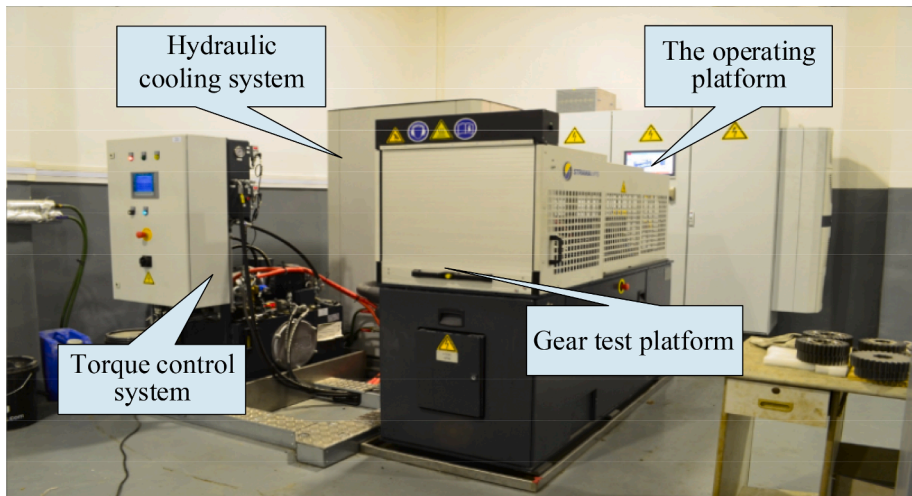


Fig. 5. Gear contact fatigue testing platform for gear dataset 2.

Table 4

Data description of gear dataset 2.

Operating condition	Dataset 1	Dataset 2	Dataset 3
Rotating speed (rpm)	1000	1000	1000
Torque (Nm)	1300	1300	1300
The number of datasets	600	600	600

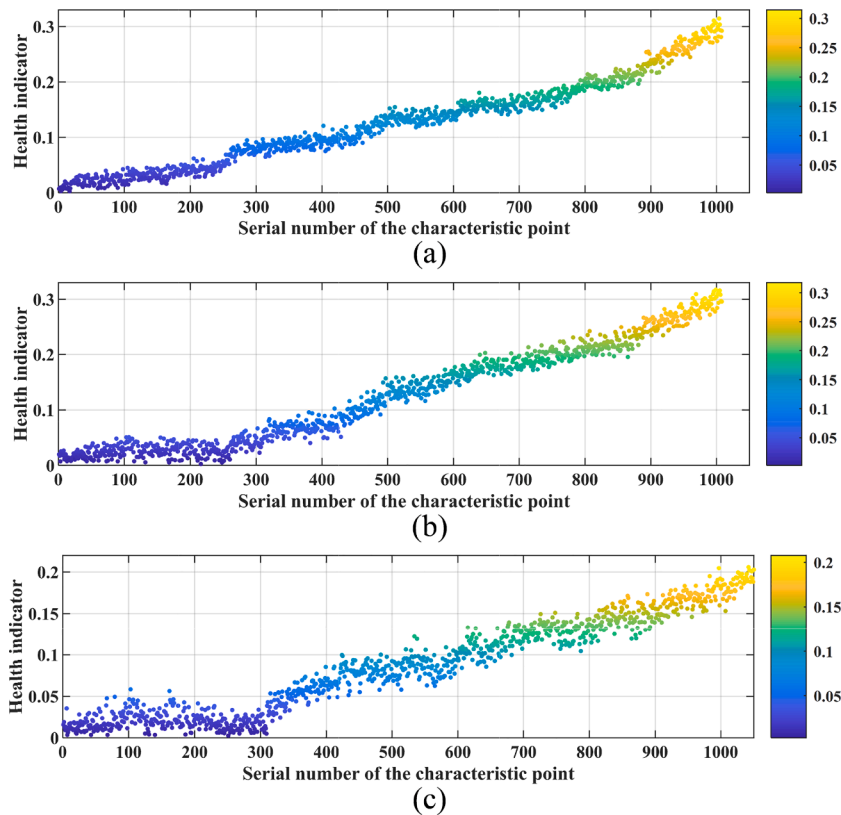


Fig. 6. Constructed gear HIs of gear dataset 1. (a) GearA1-1, (b) GearA1-2, (c) GearB1-2.

where dH_T is derivative of H_T , M is the number of the constructed HI vector, $\bar{H} = \frac{1}{M} \sum_{m=1}^M H(t_m)$, and $\bar{T} = \frac{1}{M} \sum_{m=1}^M t_m$. The parameter λ is a relaxing factor for controlling the strictness of robustness, and it is set as two.

Finally, the comprehensive performance of the gear HI is assessed though a composite index (CI), which combines the aforementioned three metrics. The CI is defined as follows:

$$CI = \theta_1 * Mon + \theta_2 * Cor + \theta_3 * Rob, \quad (30)$$

where θ_i is a positive value that satisfies $\sum_{i=1}^3 \theta_i = 1$. The parameter θ_i is determined by Ref. [34]. In this work, θ_1 , θ_2 , and θ_3 are set as 0.4, 0.3, 0.3, respectively. We think that *Mon*, *Cor*, and *Rob* play different roles in evaluating the performance of HI on RUL prediction. It follows that they are not considered equally important in this work. Specifically, *Mon* is more critical for the accurate RUL prediction compared to *Cor* and *Rob*. Consequently, *Mon* has been assigned the highest weight, while the *Cor* and *Rob* have been given relatively lower weights to reflect their less significance.

3.3. Gear HI construction by LNMM-based method

The life-cycle signals of GearA1-1, GearA1-2, and GearB1-2 in dataset 1 are used to construct HI and validate the efficacy and feasibility of the developed HI construction approach by constructing the LNMM. In the constructed LNMM, the number of both normal and lognormal distributions is configured to one. The maximum number of iterations of the EM method is configured to be 20. In addition, the *K-means* approach is employed for initializing the model parameters of LNMM. The gear HIs are obtained based on the life-cycle data and the constructed LNMM, which are shown in Fig. 6. The HIs constructed using the developed method can provide the quantitative representation of the gear's degradation process over time. Additionally, the values of the obtained HIs almost distribute over a small interval. In the initial stage, the HI value approximates zero, illustrating that the estimated data distribution of the current data is similar to the benchmark distribution. This phenomenon demonstrates that the gear is in a normal stage, and no defects are observed on the tested gear. The HI value gradually increases with the working time, illustrating that the difference between the current estimated data distribution and the benchmark distribution becomes larger. Consequently, different types of defects will occur on the tested gear.

Gear dataset 2 is also applied to confirm the effectiveness and feasibility of the LNMM-based approach. The generated gear's HIs are shown in Fig. 7. Similarly, this figure demonstrates the appropriateness of the obtained HI in representing the gear's health condition over time. Finally, according to Figs. 6 and 7, the developed HI construction method is effective and feasible in generating HI and

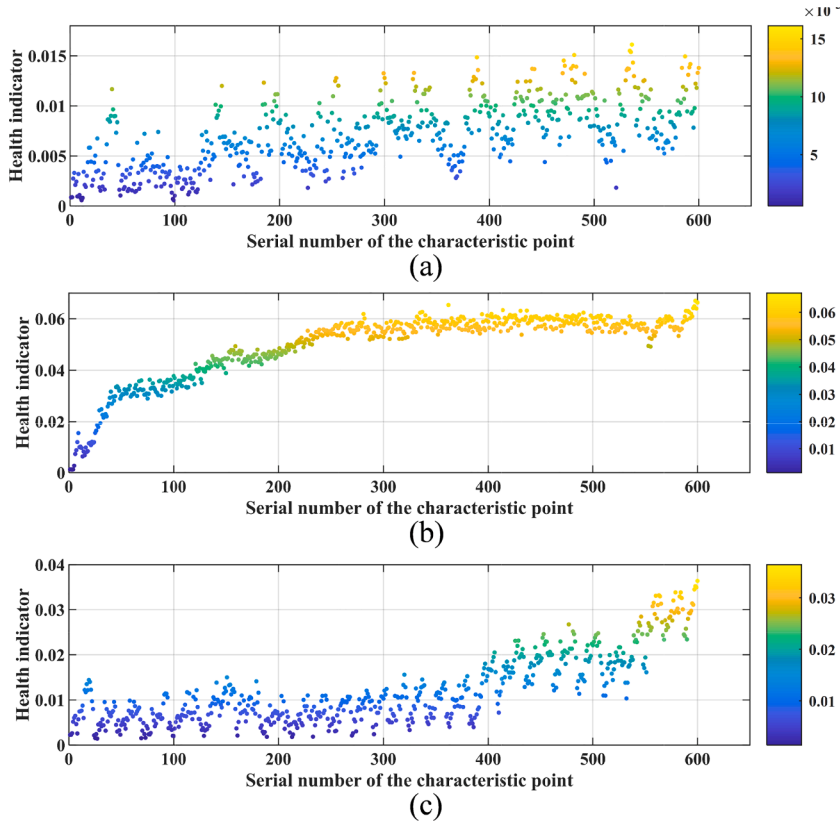


Fig. 7. Constructed gear HIs of gear dataset 2. (a) Dataset 1, (b) Dataset 2, (c) Dataset 3.

identifying the deterioration process of gear.

3.4. Comparative analysis of HI construction approaches

To demonstrate the advantage of the developed LNMM approach, a comparative experiment is conducted with the RMS, principal components analysis (PCA), kernel principal components analysis (KPCA), and isometric mapping (ISOMAP) [33] methods. Additionally, the advanced HI construction methods including GMM with Kullback-Leibler divergence (GKLD) [35] and GMM with confidence value (GCV) [36] and the HI construction methods based on SMM, GMM [28], and LMM are also employed for further comparison. For PCA, KPCA, and ISOMAP, several time-domain and frequency-domain features are employed for constructing a virtual HI by fusing these features. The evaluation metrics, such as *Mon*, *Cor*, *Rob*, and *CI*, are employed to quantifiably assess the performance of constructed HIs based on gear datasets 1 and 2. Table 5 displays the evaluation results for various methods.

According to Table 5, the LNMM has the highest comprehensive performance among these methods. Taking GearA1-1 as an example, the *CI* of the LNMM is 0.865, which is higher than those of RMS, PCA, KPCA, ISOMAP, GKLD, GCV, SMM, GMM, and LMM. For GearB1-2 and Dataset 2, the LNMM is inferior to RMS or ISOMAP, but it outperforms other methods. For Dataset 3, the *CI* of the developed LNMM is lower than those of RMS, PCA, KPCA, ISOMAP, and GKLD, but it is higher than those of GCV, SMM, GMM, and LMM. It can be also known from Table 5 that the monotonic tendency of LNMM-based HI is better than these comparative methods. Taking GearA1-1 as an example, the *Mon* of LNMM is 0.712, which is higher than those of ISOMAP, KPCA, PCA, and RMS. This indicates that the HI generated by LNMM is able to evaluate the degeneration process of gear, and the LNMM-based HI has better performance than fusion HIs. In addition, the *Mon* of LNMM is higher than those of GKLD, GCV, SMM, LMM and GMM, illustrating that the LNMM which can estimate data distributions from two data domains is better suitable for constructing gear HI than GKLD, GCV, SMM, LMM and GMM. This is because that the LNMM can extract the additional degradation characteristics from the raw data and the exponentially transformed data. The similar results can be concluded from the *Mon* results of GearA1-2, GearB1-2, Dataset 1. For *Cor* and *Rob*, the LNMM is superior to or comparable to other comparative methods, demonstrating that the LNMM-based HI has a higher temporal correlation and smaller overall volatility. Thus, the prediction accuracy can be improved when the proposed HI is employed for time-series-based life prediction.

In order to further demonstrate the superiority of the LNMM, the average evaluation results for various methods are also shown in Table 6. According to Table 6, the LNMM has the highest *CI* among these HI construction methods. the *Mon* of LNMM is 0.470, which is higher than those of RMS, PCA, KPCA, ISOMAP, GKLD, GCV, SMM, GMM, and LMM. The similar results can be concluded from the *Cor* and *Rob* values of these methods. Therefore, it can be concluded that the LNMM-based HI better assesses the degradation process of gear and has a stronger monitoring capability than other HIs.

The average calculation time of different methods is obtained for assessing the computational efficiency and computational cost of the proposed LNMM method. Taking GearA1-2 as an example, the calculation time of various methods is listed in Table 7. It can be known from Table 7 that LNMM has a longer calculation time than RMS, PCA, KPCA, and ISOMAP, while it has a shorter calculation time than GKLD, GCV, SMM, GMM, and LMM. The average calculation time of LNMM is 12.590 s, which is less than the sampling

Table 5
The evaluation results for various methods.

Method	GearA1-1				GearA1-2				GearB1-2			
	<i>Mon</i>	<i>Cor</i>	<i>Rob</i>	<i>CI</i>	<i>Mon</i>	<i>Cor</i>	<i>Rob</i>	<i>CI</i>	<i>Mon</i>	<i>Cor</i>	<i>Rob</i>	<i>CI</i>
RMS	0.678	0.963	0.956	0.847	0.605	0.978	0.949	0.820	0.445	0.972	0.930	0.749
PCA	0.071	0.717	0.812	0.487	0.200	0.843	0.800	0.573	0.240	0.907	0.851	0.623
KPCA	0.078	0.723	0.815	0.493	0.194	0.846	0.801	0.572	0.244	0.910	0.854	0.627
ISOMAP	0.221	0.884	0.835	0.604	0.313	0.925	0.860	0.661	0.283	0.925	0.864	0.650
GKLD	0.706	0.929	0.958	0.848	0.636	0.941	0.950	0.822	0.430	0.936	0.926	0.731
GCV	0.543	0.918	0.942	0.775	0.557	0.938	0.938	0.786	0.396	0.940	0.903	0.711
SMM	0.633	0.977	0.945	0.830	0.627	0.974	0.935	0.823	0.394	0.968	0.913	0.722
GMM	0.646	0.976	0.950	0.836	0.636	0.976	0.936	0.828	0.415	0.966	0.911	0.729
LMM	0.647	0.976	0.949	0.836	0.637	0.975	0.936	0.828	0.417	0.967	0.911	0.730
LNMM	0.712	0.981	0.954	0.865	0.652	0.976	0.940	0.836	0.451	0.967	0.919	0.746

Method	Dataset 1				Dataset 2				Dataset 3			
	<i>Mon</i>	<i>Cor</i>	<i>Rob</i>	<i>CI</i>	<i>Mon</i>	<i>Cor</i>	<i>Rob</i>	<i>CI</i>	<i>Mon</i>	<i>Cor</i>	<i>Rob</i>	<i>CI</i>
RMS	0.022	0.361	0.513	0.271	0.462	0.838	0.947	0.720	0.239	0.780	0.833	0.579
PCA	0.105	0.246	0.456	0.253	0.462	0.934	0.861	0.723	0.275	0.762	0.869	0.599
KPCA	0.108	0.248	0.452	0.253	0.449	0.936	0.867	0.720	0.275	0.768	0.868	0.601
ISOMAP	0.092	0.257	0.449	0.249	0.512	0.956	0.911	0.765	0.359	0.762	0.877	0.635
GKLD	0.262	0.450	0.553	0.406	0.472	0.840	0.931	0.720	0.242	0.742	0.896	0.588
GCV	0.269	0.541	0.574	0.442	0.299	0.624	0.862	0.565	0.165	0.724	0.817	0.528
SMM	0.229	0.682	0.653	0.492	0.429	0.775	0.914	0.678	0.185	0.777	0.762	0.536
GMM	0.275	0.577	0.624	0.470	0.352	0.789	0.935	0.658	0.172	0.768	0.806	0.541
LMM	0.276	0.577	0.624	0.471	0.352	0.789	0.936	0.658	0.172	0.768	0.807	0.541
LNMM	0.306	0.649	0.648	0.512	0.486	0.831	0.947	0.728	0.212	0.798	0.835	0.574

Table 6

The average evaluation results for various methods.

Method	<i>Mon</i>	<i>Cor</i>	<i>Rob</i>	<i>CI</i>
RMS	0.408	0.815	0.855	0.664
PCA	0.226	0.735	0.775	0.543
KPCA	0.225	0.738	0.776	0.544
ISOMAP	0.297	0.785	0.799	0.594
GKLD	0.458	0.806	0.869	0.686
GCV	0.371	0.781	0.839	0.634
SMM	0.416	0.859	0.854	0.680
GMM	0.416	0.842	0.860	0.677
LMM	0.417	0.842	0.861	0.678
LNMM	0.470	0.867	0.874	0.710

Table 7

The calculation time of various methods.

Method	RMS	PCA	KPCA	ISOMAP	GKLD	GCV	SMM	GMM	LMM	LNMM
Calculation time (s)	0.549	0.441	0.456	0.832	19.463	19.436	62.636	19.480	32.035	12.590

interval (60 s). Therefore, the calculation time of the proposed LNMM is sufficient for the HI construction in real-time applications. Certainly, the calculation time of HI construction can be dramatically reduced by upgrading hardware devices and optimizing device configurations when the computational resources are limited.

3.5. Gear RUL prediction with the LNMM-based HI

The generated LNMM-based HIs are employed to predict the gear's RULs via the constructed IMLMA-Net. Specifically, for the HIs of GearA1-1 and GearA1-2, the initial 948 HI values are used to construct a training dataset for predicting the remaining 60 unknown HI values. For the HI of GearB1-2, the first 1004 HI values are utilized to generate a training dataset for estimating the last 60 unknown HI values. For the HIs of Dataset 1 and Dataset 3, the first 540 HI values are employed to train the IMLMA-Net, and the last 60 HI values are estimated by the trained IMLMA-Net. Thus, the actual RULs of gears are both equal to 60 min. Since the HI of Dataset 2 has no obvious degradation trend in the last stage, it is not applied to RUL prediction in this section. Table 8 provides the hyper-parameters of IMLMA-Net for different datasets.

Using the configured hyper-parameters and training datasets, the developed IMLMA-Net is trained to obtain optimized weight parameters. Based on the trained IMLMA-Net model, the unknown HIs can be predicted step by step, and the gear RUL is computed when the estimated HI values reach the predetermined failure threshold. To guarantee reliability and fairness in the RUL prediction process, the trained IMLMA-Net is executed 100 times on each gear HI. The mean value of these estimated RULs is considered as final estimated RUL. Additionally, during the testing phase, the dropout layer in IMLMA-Net is enabled to quantify the uncertainty of the prediction model. Thus, the probability density functions (PDFs) of predicted RULs can be obtained. Table 9 and Fig. 8 display the PDFs, interval estimations, actual RULs, and predicted RULs for Gear A1-1, GearA1-2, GearB1-2, Dataset 1, and Dataset 3.

Based on the information presented in Table 9 and Fig. 8, it can be known that the predicted RULs closely align with the actual RULs. Further, the PDFs exhibit a high concentration around the predicted RULs, indicating small uncertainty. Consequently, the predictive maintenance strategies can be made based on the level of uncertainty to prevent catastrophic accidents. The PDF with minimal dispersion shows that the IMLMA-Net is confident for RUL estimation based on the LNMM-based HIs. This observation also indicates that the HIs constructed by LNMM are more suitable for RUL prediction.

In order to further demonstrate the validity of LNMM-based HI in RUL prediction, the graphical form of RUL prediction for GearA1-2 is displayed in Fig. 9. It can be known from Fig. 9 that the training and predictive curves of GearA1-2 are approximate to the corresponding actual curve. This indicates that the IMLMA-Net model can effectively learn the degradation information from the LNMM-based HI. The favourable degradation trend of LNMM-based HI can improve the accuracy of RUL prediction. To sum up, the developed LNMM-based HI is appropriate for RUL prediction of gear and is able to obtain accurate RUL prediction results.

3.6. Comparison with other HIs for RUL prediction

The constructed HIs of GearA1-2 and Dataset 1 based on RMS, PCA, KPCA, ISOMAP, GKLD, GCV, SMM, GMM, and LMM are all applied for comparison to illustrate the superiority of the LNMM-based HI. Five evaluation metrics, namely mean absolute error (MAE), normalized root mean square error (NRMSE), mean absolute percentage error (MAPE), root mean square error (RMSE), and score, are employed to quantitatively assess the predictive performance of different gear HIs. The comparison results of these five evaluation metrics for GearA1-2 and Dataset 1 are presented in Fig. 10.

According to Fig. 10, the LNMM-based HI (LNMM-HI) exhibits the best score and lowest prediction error compared to other approaches. Particularly, LNMM-HI significantly outperforms ISOMAP-based HI (ISOMAP-HI), KPCA-based HI (KPCA-HI), and PCA-

Table 8

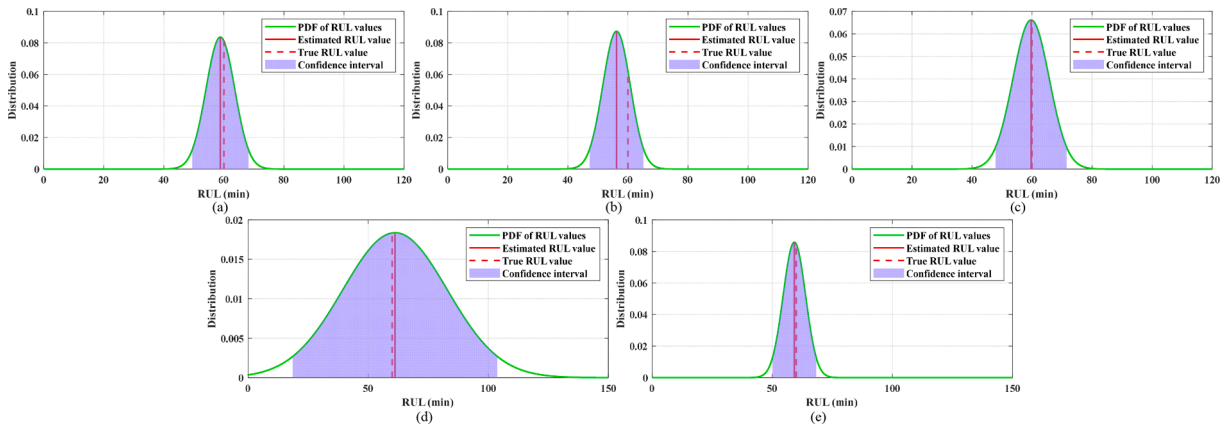
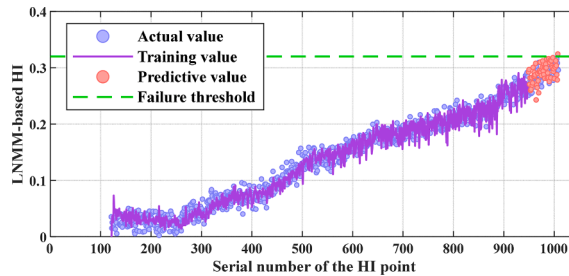
The hyper-parameters of IMLMA-Net.

Hyper-parameters	GearA1-1	GearA1-2	GearB1-2	Dataset 1	Dataset 3
Input data dimension	125	120	160	80	160
IMLMA-Net cell dimension	21	21	22	19	22
Output data dimension	1	1	1	1	1
Learning rate	0.04	0.04	0.01	0.01	0.08
Dropout rate	0.07	0.07	0.1	0.09	0.09
Optimizer	SGD	SGD	SGD	SGD	SGD
Training epoch	300	300	300	300	300

Table 9

RUL prediction with uncertainty for gears.

Gear	Actual RUL (min)	Predicted RUL (min)	95 % confidence interval
GearA1-1	60	58.84	[49.48 68.19]
GearA1-2	60	56.20	[47.24 65.16]
GearB1-2	60	59.67	[47.84 71.50]
Dataset 1	60	61.19	[18.62 103.76]
Dataset 3	60	59.20	[50.09 68.31]

**Fig. 8.** Gear's estimated results with quantified uncertainty: (a) GearA1-1, (b) GearA1-2, (c) GearB1-2, (d) Dataset 1, (e) Dataset 3.**Fig. 9.** Actual, training, and predictive curves for 60 predicted points of GearA1-2.

based HI (PCA-HI) in terms of MAE, NRMSE, MAPE, RMSE, and score. This observation demonstrates that LNMM has better performance than feature fusion methods. Moreover, LNMM-HI is better than LMM-based HI (LMM-HI), GMM-based HI (GMM-HI), SMM-based HI (SMM-HI), GCV-based HI (GCV-HI), GKLD-based HI (GKLD-HI), and RMS in terms of five evaluation metrics, demonstrating that estimating data distributions from two data domains via LNMM can generate better gear's HI for RUL prediction.

The MLMA-Net is employed for estimating RUL using LNMM-HIs of GearA1-2 and Dataset 1 to further illustrate the superiority of LNMM-HI and IMLMA-Net. Table 10 shows the evaluation results of MLMA-Net and IMLMA-Net. It can be noted that the IMLMA-Net has better prediction performance than MLMA-Net. This is because the Gumbel function in IMLMA-Net can generate more gated output values around boundary regions compared to the MAF in MLMA-Net. According to the comparison of the evaluation results of MLMA-

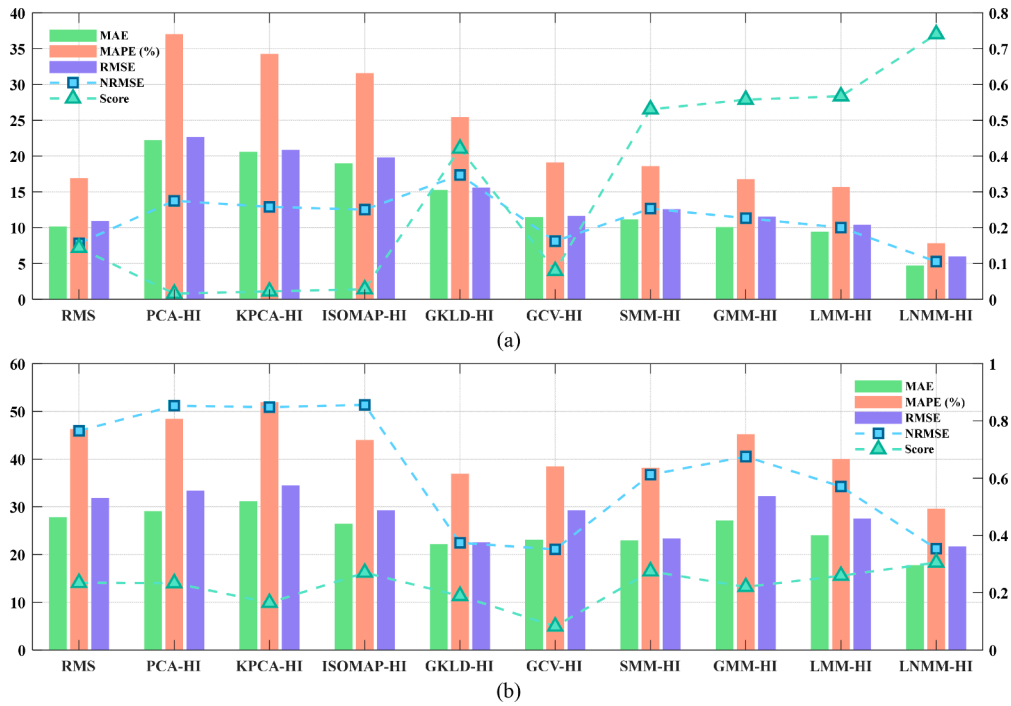


Fig. 10. The comparison results of evaluation metrics: (a) GearA1-2, (b) Dataset 1.

Table 10

The evaluation results for MLMA-Net and IMLMA-Net.

	Methods	MAE	NRMSE	MAPE	RMSE	Score
GearA1-2	MLMA-Net	9.4200	0.1715	15.70 %	11.7762	0.3187
	IMLMA-Net	4.6600	0.1054	7.77 %	5.9262	0.7405
Dataset 1	MLMA-Net	22.1900	0.4222	36.98 %	26.6786	0.2374
	IMLMA-Net	17.7100	0.3537	29.52 %	21.6423	0.3053

Net based on LNMM-HI with the evaluation results of IMLMA-Net based on other HIs, it can be observed that the LNMM-HI is more appropriate to RUL prediction. In conclusion, the developed LNMM-based HI demonstrates superior prediction performance compared to other HIs.

4. Conclusions

The existing HI construction methods based on GMM use a single type of distribution to obtain the distribution discrepancy from a single data domain. Evidently, the distribution characteristics in a single domain cannot comprehensively reveal the health condition and degeneration process of gear. Hence, the LMM was initially developed to capture the degradation characteristics within the exponential data domain and fill this gap while mining additional degradation information from different domains. Then, a novel mixture model named LNMM was proposed based on the GMM and LMM. The LNMM can extract normal and lognormal distribution characteristics from two data domains, including raw vibration data and exponentially transformed data. Then, the distribution discrepancies between the benchmark distribution of health signal and distributions of the full life-cycle signals were obtained using the DCRM. Next, the gear's HI was constructed by combining the DCRMs of the two domains. The availability and feasibility of the developed HI construction approach were verified by several gear life-cycle datasets. The gear's RULs are accurately predicted by the IMLMA-Net and LNMM-based HI. Moreover, the uncertainty of IMLMA-Net was quantified via the PDFs of the estimated RULs. Finally, the comparative results illustrate the superiority of the developed HI construction approach over other unsupervised methodologies. Additionally, the constructed HI demonstrates notable potential for accurate life prediction of gear.

The proposed LNMM-based HI construction method successfully constructed gear HIs under stationary operation conditions. Under time-varying working conditions, the distribution characteristics of monitoring data not only change with the degradation process, but also with the varied working conditions, so that the HI constructed by the proposed method cannot accurately describe the real degradation process. In this case, the proposed HI would lead to the inaccurate RUL prediction results. Therefore, the further work

needs to study the influence of time-varying working condition on the distribution of acquired monitoring data, so as to explore an unsupervised HI construction method based on mixture model under non-stationary operational conditions.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

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